Inverse Function Theorem

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In multivariable calculus, the Inverse Function Theorem provides the mathematical conditions under which a complex system can be reverse-engineered. It relies heavily on the properties of the Jacobian matrix.

Theorem 1 (Multivariable Inverse Function Theorem)**.** *Let* $\mathbf{f} : \mathbb{R}^n \to \mathbb{R}^n$ *be a continuously differentiable function defined on an open set containing a point* $\mathbf{a}$*. Let* $\mathbf{b} = \mathbf{f}(\mathbf{a})$*.*

If the Jacobian matrix of $\mathbf{f}$ *at point* $\mathbf{a}$*, denoted as* $J_{\mathbf{f}}(\mathbf{a})$*, is invertible—meaning its determinant is non-zero:*

$$\det(J_{\mathbf{f}}(\mathbf{a})) \neq 0$$

Then, the following three conditions hold true:

1. ***Local Invertibility:*** *There exists an open neighborhood* U *around* $\mathbf{a}$ *and an open neighborhood* V *around* $\mathbf{b}$ *such that* $\mathbf{f}$ *is a bijection from* U *to* V*. Within this local space, the inverse function* $\mathbf{f}^{-1}$ *strictly exists.*

2. ***Differentiability of the Inverse:*** *The inverse function* $\mathbf{f}^{-1}$ *is continuously differentiable on* V*.*

3. ***The Inverse Jacobian:*** *The Jacobian matrix of the inverse function at* $\mathbf{b}$ *is exactly equal to the mathematical matrix inverse of the Jacobian of the original function at* $\mathbf{a}$*. Formally:*

$$J_{\mathbf{f}^{-1}}(\mathbf{b}) = [J_{\mathbf{f}}(\mathbf{a})]^{-1}$$

Proof of Point 1: Local Invertibility. **Step 1: Setting up the problem.**

We want to prove that, near point $\mathbf{a}$, the equation $\mathbf{f}(\mathbf{x}) = \mathbf{y}$ has exactly one solution $\mathbf{x}$ for every point $\mathbf{y}$. We know that the theorem states that $\mathbf{det}(\mathbf{A}) \neq \mathbf{0} \to \exists A^{-1}$.

Let us $\mathbf{A}$ represent the Jacobian matrix evaluated at point a:

$$A = J_f(a)$$

We want to know how to solve $\mathbf{f}(\mathbf{x}) = \mathbf{y}$. Multiply both sides by $\mathbf{A^{-1}}$:

$$A^{-1}f(x) = A^{-1}y$$

Then rearrange:

$$A^{-1}(f(x) - y) = 0$$

$$x - A^{-1}(f(x) - y) = x$$

Step 2: Creating a new function.

Let us define the left-hand side of the equation above as a new function:

$$\Phi_y(x) = x - A^{-1}(f(x) - y)$$

And note that solving $\mathbf{f}(\mathbf{x}) = \mathbf{y}$ is exactly the same as finding a fixed point for $\Phi_y(x) = x$.

Step 3: Analyzing the derivative (Jacobian).

Let us find a Jacobian matrix for $\mathbf{\Phi_y}(\mathbf{x})$ with respect to $\mathbf{x}$.
We will use the rules of Matrix differentiation. We take the derivative respect to $\mathbf{x}$.

1. The derivative of the identity mapping $\mathbf{x}$ is the Identity Matrix, $\mathbf{I}$.
2. Term 2 $(\mathbf{A^{-1}}(\mathbf{f}(\mathbf{x}) - \mathbf{y})) : \mathbf{A^{-1}}$ is a constant matrix, so it stays out front.
   (a) The derivative of $\mathbf{f}(\mathbf{x})$ is $\mathbf{J_f}(\mathbf{x})$.
   (b) The derivative of the constant vector $\mathbf{y}$ is 0.

Putting it together:

$$J_{\Phi(x)} = I - A^{-1} J_f(x)$$

Now evaluate our this derivative specially at point a:

$$J_{\Phi(x)}(a) = I - A^{-1} J_f(a)$$

but

$$J_f(a) = A$$

So

$$J_\Phi(a) = I - A^{-1}A = I - I = 0$$

Step 4: The Contraction Mapping.
We know that $\mathbf{f} \in \mathbf{C^1}$ if and only if all of its component functions $(f_1, \ldots, f_n)$ have continuous partial derivatives.

$$J_{\mathbf{f}}(\mathbf{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{pmatrix}$$

If we assume $\mathbf{f}$ to be C^1, we explicitly state that every cell in that matrix is a continuous function of $\mathbf{x}$.
Let us consider the **Frobenius norm**:

$$\|J_{\Phi_y}(\mathbf{x})\|_F = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} |g_{ij}(\mathbf{x})|^2}$$

Notice:

1. If $g_{ij}(\mathbf{x})$ is continuous, then its square $(g_{ij}(\mathbf{x}))^2$ is continuous.
2. The sum of continuous functions is continuous.
3. The square root of a continuous function is continuous (where the inside is non-negative).

That means that the mapping $\mathbf{x} \mapsto \|J_{\Phi_y}(\mathbf{x})\|_F$ is continuous.
We know from **Step 3** that $J_{\Phi_y}(\mathbf{a}) = 0$, and therefore $\|J_{\Phi_y}(\mathbf{a})\|_F = 0$.
Because $h(\mathbf{x}) = \|J_{\Phi_y}(\mathbf{x})\|_F$ is continuous, we can apply the ϵ-δ definition of continuity. If we choose an error of $\epsilon = \frac{1}{2}$, there must exist some radius $r_0 > 0$ and $r < r_0$ such that:

$$\|\mathbf{x} - \mathbf{a}\| \leq r < r_0 \implies |\|J_\Phi(\mathbf{x})\| - \|J_\Phi(\mathbf{a})\|| < \frac{1}{2}$$

Substituting $\|J_\Phi(\mathbf{a})\| = 0$, we find that for all $\mathbf{x}$ in the closed ball $\overline{B_r(\mathbf{a})}$:

$$\|J_\Phi(\mathbf{x})\| < \frac{1}{2}$$

For any two points $\mathbf{x_1}, \mathbf{x_2} \in \overline{B_r(\mathbf{a})}$. Since the closed ball $\overline{B_r(\mathbf{a})}$ is convex, the line segment $[\mathbf{x_1}, \mathbf{x_2}]$ lies entirely in $\overline{B_r(\mathbf{a})}$. let's use **Multivariable Mean Value Inequality**:

$$\|\Phi_y(\mathbf{x_1}) - \Phi_y(\mathbf{x_2})\| \leq \left(\sup_{\mathbf{z} \in [\mathbf{x_1}, \mathbf{x_2}]} \|J_\Phi(\mathbf{z})\| \right) \|\mathbf{x_1} - \mathbf{x_2}\|$$

Since we proved that $\|J_\Phi(\mathbf{z})\| < \frac{1}{2}$ for every point in this ball, we conclude:

$$\|\Phi_y(\mathbf{x_1}) - \Phi_y(\mathbf{x_2})\| \leq \frac{1}{2}\|\mathbf{x_1} - \mathbf{x_2}\|$$

Step 5: Showing that Φ_y maps a closed ball into itself.

Our goal is to apply the Banach Fixed-Point Theorem. For that, it is not enough to know that Φ_y is a contraction; we must also show that it sends a complete metric space into itself.

So we now want to prove that there exists a closed ball centered at $\mathbf{a}$ such that

$$\Phi_y(X) \subseteq X.$$

We define

$$X = \overline{B_r(\mathbf{a})},$$

the closed ball of radius r centered at $\mathbf{a}$. Since X is a closed subset of $\mathbb{R}^n$, and $\mathbb{R}^n$ is complete, the space X is also complete.

Next, let

$$\mathbf{b} = \mathbf{f}(\mathbf{a}).$$

In the future, we want to solve the equation

$$\mathbf{f}(\mathbf{x}) = \mathbf{y}$$

for $\mathbf{y}$ close to $\mathbf{b}$. For that reason, we restrict $\mathbf{y}$ to a small neighborhood of $\mathbf{b}$.

Choose $\delta > 0$ such that

$$\|A^{-1}\|\delta \leq \frac{r}{2},$$

and define

$$V = B_\delta(\mathbf{b}).$$

Now fix any $\mathbf{y} \in V$. We want to prove that for every $\mathbf{x} \in X$, the point $\Phi_y(\mathbf{x})$ also belongs to X. By definition of X, this means that we must show

$$\|\Phi_y(\mathbf{x}) - \mathbf{a}\| \leq r.$$

To estimate this quantity, we compare $\Phi_y(\mathbf{x})$ first with $\Phi_y(\mathbf{a})$, and then compare $\Phi_y(\mathbf{a})$ with $\mathbf{a}$. This is the key idea of the argument.

First, let us estimate $\Phi_y(\mathbf{a})$. By definition of Φ_y,

$$\Phi_y(\mathbf{a}) = \mathbf{a} - A^{-1}(\mathbf{f}(\mathbf{a}) - \mathbf{y}).$$

Since $\mathbf{b} = \mathbf{f}(\mathbf{a})$, this becomes

$$\Phi_y(\mathbf{a}) = \mathbf{a} - A^{-1}(\mathbf{b} - \mathbf{y}) = \mathbf{a} + A^{-1}(\mathbf{y} - \mathbf{b}).$$

Subtracting $\mathbf{a}$ from both sides, we get

$$\Phi_y(\mathbf{a}) - \mathbf{a} = A^{-1}(\mathbf{y} - \mathbf{b}).$$

Therefore, using the matrix norm inequality,

$$\|\Phi_y(\mathbf{a}) - \mathbf{a}\| = \|A^{-1}(\mathbf{y} - \mathbf{b})\| \leq \|A^{-1}\| \, \|\mathbf{y} - \mathbf{b}\|.$$

Because $\mathbf{y} \in V = B_\delta(\mathbf{b})$, we have

$$\|\mathbf{y} - \mathbf{b}\| < \delta.$$

Hence,

$$\|\Phi_y(\mathbf{a}) - \mathbf{a}\| < \|A^{-1}\|\delta \leq \frac{r}{2}.$$

So we have shown that $\Phi_y(\mathbf{a})$ lies within distance $r/2$ of the center $\mathbf{a}$.

Now let $\mathbf{x} \in X$. Since $X = \overline{B_r(\mathbf{a})}$, by definition we have

$$\|\mathbf{x} - \mathbf{a}\| \leq r.$$

From Step 4, we already know that Φ_y is a contraction on X with contraction constant $\frac{1}{2}$. Therefore,

$$\|\Phi_y(\mathbf{x}) - \Phi_y(\mathbf{a})\| \leq \frac{1}{2}\|\mathbf{x} - \mathbf{a}\|.$$

Since $\|\mathbf{x} - \mathbf{a}\| \leq r$, this gives

$$\|\Phi_y(\mathbf{x}) - \Phi_y(\mathbf{a})\| \leq \frac{r}{2}.$$

Thus, we now have two estimates:

$$\|\Phi_y(\mathbf{x}) - \Phi_y(\mathbf{a})\| \leq \frac{r}{2} \quad \text{and} \quad \|\Phi_y(\mathbf{a}) - \mathbf{a}\| \leq \frac{r}{2}.$$

We combine them using the triangle inequality:

$$\|\Phi_y(\mathbf{x}) - \mathbf{a}\| \leq \|\Phi_y(\mathbf{x}) - \Phi_y(\mathbf{a})\| + \|\Phi_y(\mathbf{a}) - \mathbf{a}\|.$$

Substituting the two bounds obtained above, we find

$$\|\Phi_y(\mathbf{x}) - \mathbf{a}\| \leq \frac{r}{2} + \frac{r}{2} = r.$$

Therefore,

$$\Phi_y(\mathbf{x}) \in \overline{B_r(\mathbf{a})} = X.$$

Since $\mathbf{x} \in X$ was arbitrary, we conclude that

$$\Phi_y(X) \subseteq X.$$

This is exactly what we needed for the next step: now Φ_y is not only a contraction, but a contraction from the complete metric space X into itself. Therefore, in the next step we may apply the Banach Fixed-Point Theorem.

Step 6: Conclusion via Banach Fixed-Point Theorem.

Definition. Let (X, d) be a metric space with metric $d(x, y)$. Then a map $T : X \to X$ is called a *contraction mapping* on X if there exists a nonnegative constant $q < 1$ such that

$$d(T(x), T(y)) \leq q d(x, y)$$

for all $x, y \in X$.

From **step 4** we know that $\Phi_y(x)$ is a contraction mapping.

Theorem 2 (Banach Fixed-Point Theorem)**.** *Let (X, d) be a non-empty complete metric space with a contraction mapping $T : X \to X$. Then T admits a unique fixed point $x^* \in X$, such that $T(x^*) = x^*$.*

Applying the **Banach Fixed-Point Theorem** to our mapping $\Phi_y(x)$:

1. **Completeness:** Since X is a closed subset of a complete metric space (e.g., $\mathbb{R}^n$), (X, d) is itself a complete metric space.

2. **Contraction:** As shown in **Step 4**, Φ_y is a contraction mapping with constant $q < 1$.

Therefore, there exists a **unique** fixed point $x^* \in X$ such that:

$$\Phi_y(x^*) = x^*$$

By the definition of our mapping Φ_y, this fixed point x^* is the unique solution to the equation $\Phi_y(x) = x - A^{-1}(f(x) - y)$ within the neighborhood X. This completes the proof of existence and uniqueness.

Thus, for every $\mathbf{y} \in V$, the Banach Fixed-Point Theorem gives a unique point $\mathbf{x}^* \in X$ such that

$$\Phi_y(\mathbf{x}^*) = \mathbf{x}^*.$$

By the definition of Φ_y, this is equivalent to

$$\mathbf{f}(\mathbf{x}^*) = \mathbf{y}.$$

Hence, for every $\mathbf{y} \in V$, there exists a unique point $\mathbf{x}^* \in X$ such that

$$\mathbf{f}(\mathbf{x}^*) = \mathbf{y}.$$

In particular, taking $\mathbf{y} = \mathbf{b} = \mathbf{f}(\mathbf{a})$, we see that $\mathbf{a}$ is a fixed point of $\Phi_{\mathbf{b}}$, since

$$\Phi_{\mathbf{b}}(\mathbf{a}) = \mathbf{a} - A^{-1}(\mathbf{f}(\mathbf{a}) - \mathbf{b}) = \mathbf{a}.$$

By uniqueness of the fixed point in X, this implies that $\mathbf{a}$ is the unique point of X such that

$$\mathbf{f}(\mathbf{x}) = \mathbf{b}.$$

Now let

$$\partial X = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{a}\| = r\}$$

be the boundary of X. Since ∂X is compact and $\mathbf{f}$ is continuous, the set $\mathbf{f}(\partial X)$ is compact. Moreover, because $\mathbf{a}$ is the unique point of X satisfying $\mathbf{f}(\mathbf{x}) = \mathbf{b}$, and $\mathbf{a}$ lies in the interior of X, we must have

$$\mathbf{b} \notin \mathbf{f}(\partial X).$$

Since $\mathbf{f}(\partial X)$ is compact and $\mathbf{b} \notin \mathbf{f}(\partial X)$, the distance from $\mathbf{b}$ to $\mathbf{f}(\partial X)$ is strictly positive. Define

$$d := \operatorname{dist}\big(\mathbf{b}, \mathbf{f}(\partial X)\big) > 0,$$

and choose $0 < \delta' < \min\{\delta, d\}$. Set

$$V' := B_{\delta'}(\mathbf{b}).$$

Then V' is an open neighborhood of $\mathbf{b}$ contained in V, and by construction

$$V' \cap \mathbf{f}(\partial X) = \varnothing.$$

We now define

$$U := B_r(\mathbf{a}) \cap \mathbf{f}^{-1}(V').$$

Since both $B_r(\mathbf{a})$ and $\mathbf{f}^{-1}(V')$ are open, U is open. Also, $\mathbf{a} \in U$ because $\mathbf{f}(\mathbf{a}) = \mathbf{b} \in V'$.

We claim that

$$\mathbf{f} : U \to V'$$

is bijective.

First, let $\mathbf{y} \in V'$. Since $V' \subseteq V$, the previous argument gives a unique point $\mathbf{x}^* \in X$ such that

$$\mathbf{f}(\mathbf{x}^*) = \mathbf{y}.$$

We show that in fact $\mathbf{x}^* \in B_r(\mathbf{a})$. Indeed, if $\mathbf{x}^* \in \partial X$, then

$$\mathbf{y} = \mathbf{f}(\mathbf{x}^*) \in \mathbf{f}(\partial X),$$

which contradicts $V' \cap \mathbf{f}(\partial X) = \varnothing$. Hence $\mathbf{x}^*$ lies in the interior of X, that is,

$$\mathbf{x}^* \in B_r(\mathbf{a}).$$

Since also $\mathbf{f}(\mathbf{x}^*) = \mathbf{y} \in V'$, we have $\mathbf{x}^* \in U$. Therefore, $\mathbf{f} : U \to V'$ is surjective.

Next, we prove injectivity. Let $\mathbf{x}_1, \mathbf{x}_2 \in U$ and suppose that

$$\mathbf{f}(\mathbf{x}_1) = \mathbf{f}(\mathbf{x}_2) = \mathbf{y}$$

for some $\mathbf{y} \in V'$. Because $U \subset B_r(\mathbf{a}) \subset X$, both $\mathbf{x}_1$ and $\mathbf{x}_2$ belong to X. But for each $\mathbf{y} \in V \supset V'$, we already proved that there exists a unique point of X mapped to $\mathbf{y}$. Therefore,

$$\mathbf{x}_1 = \mathbf{x}_2.$$

So $\mathbf{f} : U \to V'$ is injective.

We conclude that there exist open neighborhoods U of $\mathbf{a}$ and V' of $\mathbf{b}$ such that

$$\mathbf{f} : U \to V'$$

is bijective. This proves the local invertibility of $\mathbf{f}$ near $\mathbf{a}$.

□

Proof of Point 2 and 3: Differentiability and Inverse Jacobian. **Step 1: Proving the inverse function is Lipschitz Continuous.**
Before we take the derivative, we must ensure the inverse function $\mathbf{g} = \mathbf{f}^{-1}$ must be continuous. We can prove this using the contraction mapping property from Point 1.

Recall from Point 1, Step 4 that for any $\mathbf{x_1}, \mathbf{x_2}$ in our neighborhood:

$$\|\Phi_y(\mathbf{x_1}) - \Phi_y(\mathbf{x_2})\| \leq \frac{1}{2}\|\mathbf{x_1} - \mathbf{x_2}\|$$

By the definition of Φ, this means:

$$\left\|(\mathbf{x_1} - \mathbf{x_2}) - A^{-1}(\mathbf{f}(\mathbf{x_1}) - \mathbf{f}(\mathbf{x_2}))\right\| \leq \frac{1}{2}\|\mathbf{x_1} - \mathbf{x_2}\|$$

By the reverse triangle inequality ($\|u\| - \|v\| \leq \|u - v\|$), we can write:

$$\|\mathbf{x_1} - \mathbf{x_2}\| - \|A^{-1}(\mathbf{f}(\mathbf{x_1}) - \mathbf{f}(\mathbf{x_2}))\| \leq \frac{1}{2}\|\mathbf{x_1} - \mathbf{x_2}\|$$

Rearranging this gives:

$$\frac{1}{2}\|\mathbf{x_1} - \mathbf{x_2}\| \leq \|A^{-1}\|\|\mathbf{f}(\mathbf{x_1}) - \mathbf{f}(\mathbf{x_2})\|$$

Since $\mathbf{g}(\mathbf{y}) = \mathbf{x}$, we can substitute $\mathbf{x}$ with $\mathbf{g}(\mathbf{y})$:

$$\|\mathbf{g}(\mathbf{y_1}) - \mathbf{g}(\mathbf{y_2})\| \leq 2\|A^{-1}\|\|\mathbf{y_1} - \mathbf{y_2}\|$$

Because $\|A^{-1}\|$ is just a constant, this proves $\mathbf{g}$ is Lipschitz continuous. Crucially, as $\mathbf{y} \to \mathbf{y_0}$, it is guaranteed that $\mathbf{x} \to \mathbf{x_0}$.

Step 2: Defining the derivative of the original function.

Let us use differentiability in the sense of Fréchet. Because $\mathbf{f}$ is differentiable at $\mathbf{x_0}$, we can write it using a linear approximation plus a remainder term $R(\mathbf{x})$:

$$\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x_0}) = J_{\mathbf{f}}(\mathbf{x_0})(\mathbf{x} - \mathbf{x_0}) + R(\mathbf{x})$$

By the definition of differentiability, the remainder goes to zero faster than the distance:

$$\lim_{\mathbf{x} \to \mathbf{x_0}} \frac{\|R(\mathbf{x})\|}{\|\mathbf{x} - \mathbf{x_0}\|} = 0$$

Step 3: Evaluating the limit for the inverse function.

Since $\mathbf{f} \in C^1$, the map $\mathbf{x} \mapsto J_{\mathbf{f}}(\mathbf{x})$ is continuous, and hence

$$\mathbf{x} \mapsto \det(J_{\mathbf{f}}(\mathbf{x}))$$

is continuous. Because

$$\det(J_{\mathbf{f}}(\mathbf{a})) \neq 0,$$

there exists an open neighborhood U_0 of $\mathbf{a}$ such that

$$\det(J_{\mathbf{f}}(\mathbf{x})) \neq 0 \quad \text{for all } \mathbf{x} \in U_0.$$

Let

$$U_1 := U \cap U_0.$$

Since U and U_0 are open neighborhoods of $\mathbf{a}$, the set U_1 is also an open neighborhood of $\mathbf{a}$. Define

$$V_1 := \mathbf{f}(U_1) \subseteq V'.$$

Since $\mathbf{f} : U \to V'$ is bijective, its restriction

$$\mathbf{f} : U_1 \to V_1$$

is also bijective. Let

$$\mathbf{g}_1 : V_1 \to U_1$$

denote the inverse of this restricted map.

Therefore, for every $\mathbf{y}_0 \in V_1$, if we set

$$\mathbf{x}_0 = \mathbf{g}_1(\mathbf{y}_0),$$

then

$$\mathbf{x}_0 \in U_1 \subseteq U_0.$$

Hence $J_{\mathbf{f}}(\mathbf{x}_0)$ is invertible, so the matrix

$$[J_{\mathbf{f}}(\mathbf{x}_0)]^{-1}$$

is well-defined.

Let $\mathbf{y_0} \in V_1$ and $\mathbf{x_0} = \mathbf{g_1}(\mathbf{y_0})$. We hypothesize that the Jacobian of the inverse is the inverse of the Jacobian: $J_{\mathbf{g_1}}(\mathbf{y_0}) = [J_{\mathbf{f}}(\mathbf{x_0})]^{-1}$. But we must prove that $det(J_f(x_0)) \neq 0$.

To prove this, we must show that the following limit equals 0:

$$L = \lim_{\mathbf{y} \to \mathbf{y_0}} \frac{\|\mathbf{g_1}(\mathbf{y}) - \mathbf{g_1}(\mathbf{y_0}) - [J_{\mathbf{f}}(\mathbf{x_0})]^{-1}(\mathbf{y} - \mathbf{y_0})\|}{\|\mathbf{y} - \mathbf{y_0}\|}$$

Substitute $\mathbf{y} = \mathbf{f}(\mathbf{x})$ and $\mathbf{g_1}(\mathbf{y}) = \mathbf{x}$. Because $\mathbf{g_1}$ is continuous (from Step 1), as $\mathbf{y} \to \mathbf{y_0}$, we can change the limit to $\mathbf{x} \to \mathbf{x_0}$:

$$L = \lim_{\mathbf{x} \to \mathbf{x_0}} \frac{\|\mathbf{x} - \mathbf{x_0} - [J_{\mathbf{f}}(\mathbf{x_0})]^{-1}(\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x_0}))\|}{\|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x_0})\|}$$

Substitute the linear approximation of $\mathbf{f}$ from Step 2 into the numerator:

$$L = \lim_{\mathbf{x} \to \mathbf{x_0}} \frac{\|\mathbf{x} - \mathbf{x_0} - [J_{\mathbf{f}}(\mathbf{x_0})]^{-1}\big(J_{\mathbf{f}}(\mathbf{x_0})(\mathbf{x} - \mathbf{x_0}) + R(\mathbf{x})\big)\|}{\|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x_0})\|}$$

Distribute the inverse matrix. The term $[J_{\mathbf{f}}(\mathbf{x_0})]^{-1} J_{\mathbf{f}}(\mathbf{x_0})$ becomes the Identity matrix, leaving just $(\mathbf{x} - \mathbf{x_0})$, which cancels out with the first part of the numerator:

$$L = \lim_{\mathbf{x} \to \mathbf{x_0}} \frac{\| - [J_{\mathbf{f}}(\mathbf{x_0})]^{-1} R(\mathbf{x})\|}{\|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x_0})\|}$$

Using matrix norm properties ($\|AB\| \leq \|A\|\|B\|$):

$$L \leq \|[J_{\mathbf{f}}(\mathbf{x_0})]^{-1}\| \lim_{\mathbf{x} \to \mathbf{x_0}} \frac{\|R(\mathbf{x})\|}{\|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x_0})\|}$$

Multiply the top and bottom of the fraction by $\|\mathbf{x} - \mathbf{x_0}\|$:

$$L \leq \|[J_{\mathbf{f}}(\mathbf{x_0})]^{-1}\| \lim_{\mathbf{x} \to \mathbf{x_0}} \left(\frac{\|R(\mathbf{x})\|}{\|\mathbf{x} - \mathbf{x_0}\|} \cdot \frac{\|\mathbf{x} - \mathbf{x_0}\|}{\|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x_0})\|} \right)$$

Now, evaluate the pieces as $\mathbf{x} \to \mathbf{x_0}$:

1. $\|[J_{\mathbf{f}}(\mathbf{x_0})]^{-1}\|$ is a constant.
2. $\frac{\|R(\mathbf{x})\|}{\|\mathbf{x}-\mathbf{x_0}\|} \to 0$ (from Step 2).
3. $\frac{\|\mathbf{x}-\mathbf{x_0}\|}{\|\mathbf{f}(\mathbf{x})-\mathbf{f}(\mathbf{x_0})\|}$ is bounded by $2\|A^{-1}\|$ (from the Lipschitz property in Step 1).

Because we are multiplying a constant, by zero, by a bounded number, the entire limit $L = 0$.

By definition, because this limit equals 0, the inverse function $\mathbf{g_1}$ is differentiable at $\mathbf{y_0}$, and its Jacobian is exactly $[J_{\mathbf{f}}(\mathbf{x_0})]^{-1}$.

Since $\mathbf{y}_0 \in V_1$ was arbitrary, we have shown that for every $\mathbf{y} \in V_1$,

$$J_{\mathbf{g}_1}(\mathbf{y}) = \left[J_{\mathbf{f}}(\mathbf{g}_1(\mathbf{y}))\right]^{-1}.$$

We now prove that this Jacobian depends continuously on $\mathbf{y}$.

First, $\mathbf{g}_1$ is continuous on V_1 because it is Lipschitz. Second, since $\mathbf{f} \in C^1$, the map

$$\mathbf{x} \mapsto J_{\mathbf{f}}(\mathbf{x})$$

is continuous. Third, matrix inversion

$$M \mapsto M^{-1}$$

is continuous on the set of invertible matrices. Since $J_{\mathbf{f}}(\mathbf{g}_1(\mathbf{y}))$ is invertible for every $\mathbf{y} \in V_1$, it follows that the composition

$$\mathbf{y} \mapsto J_{\mathbf{g}_1}(\mathbf{y}) = \left[J_{\mathbf{f}}(\mathbf{g}_1(\mathbf{y}))\right]^{-1}$$

is continuous on V_1.

Therefore, $\mathbf{g}_1$ is continuously differentiable on V_1, i.e.

$$\mathbf{g}_1 \in C^1(V_1).$$

This proves the differentiability of the inverse and the formula for its Jacobian. $\square$